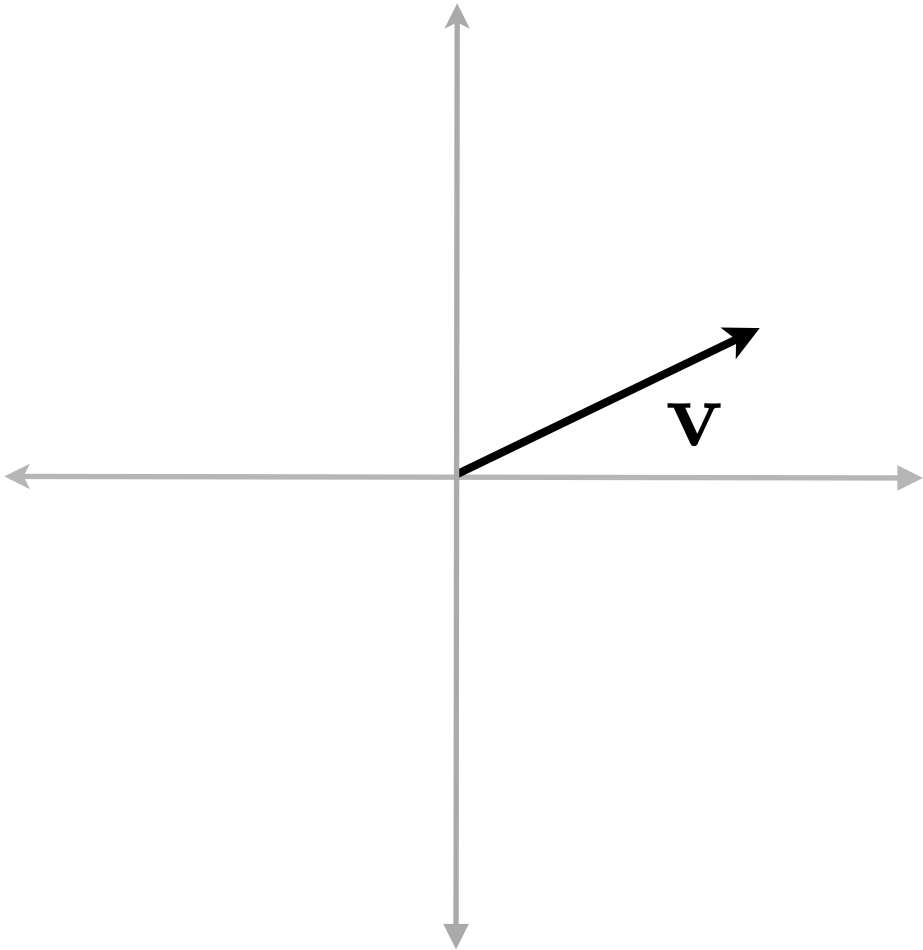
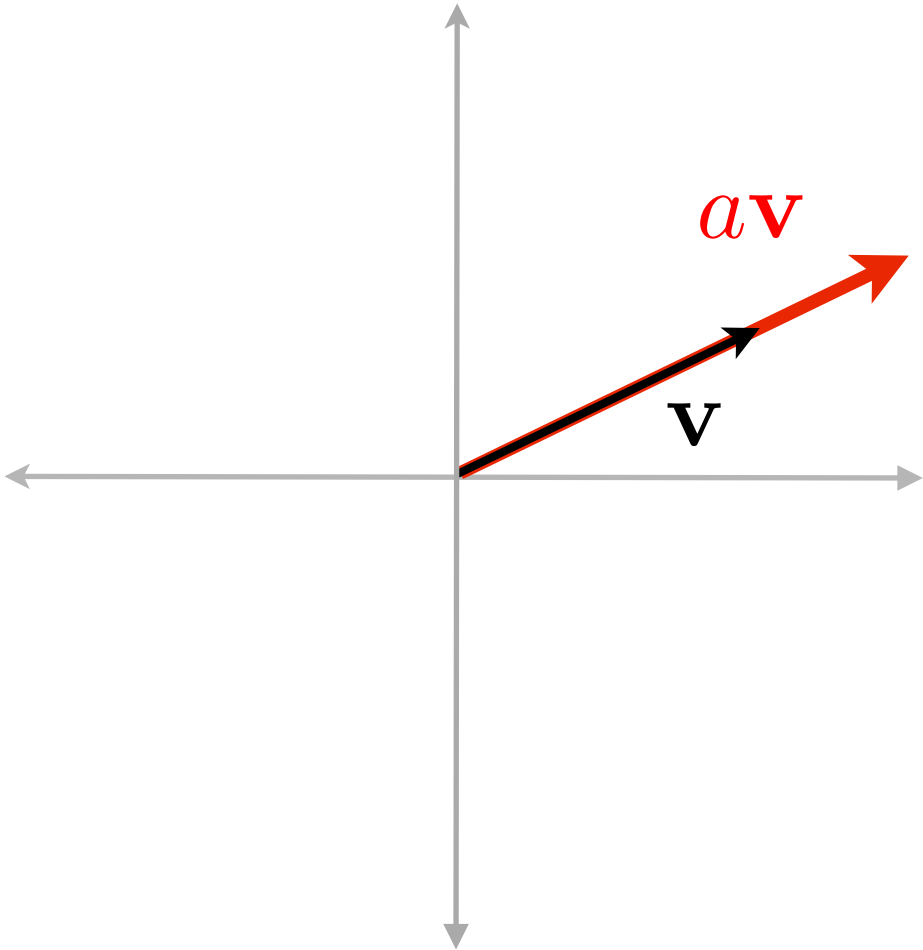
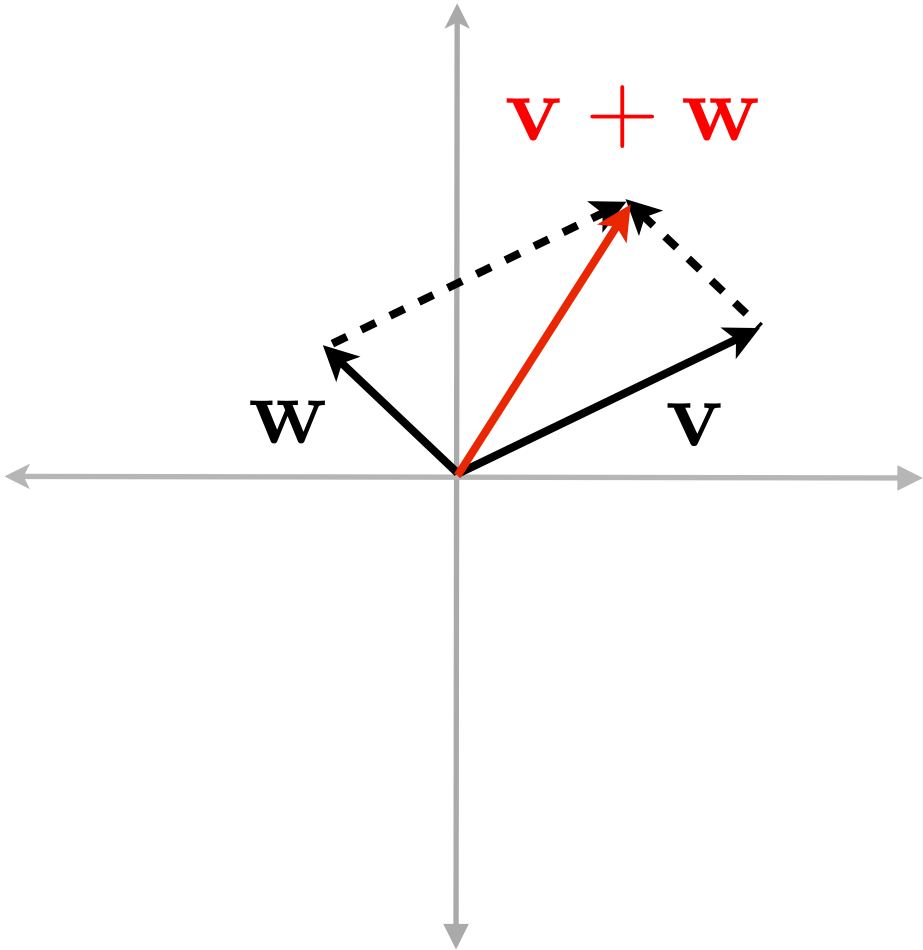
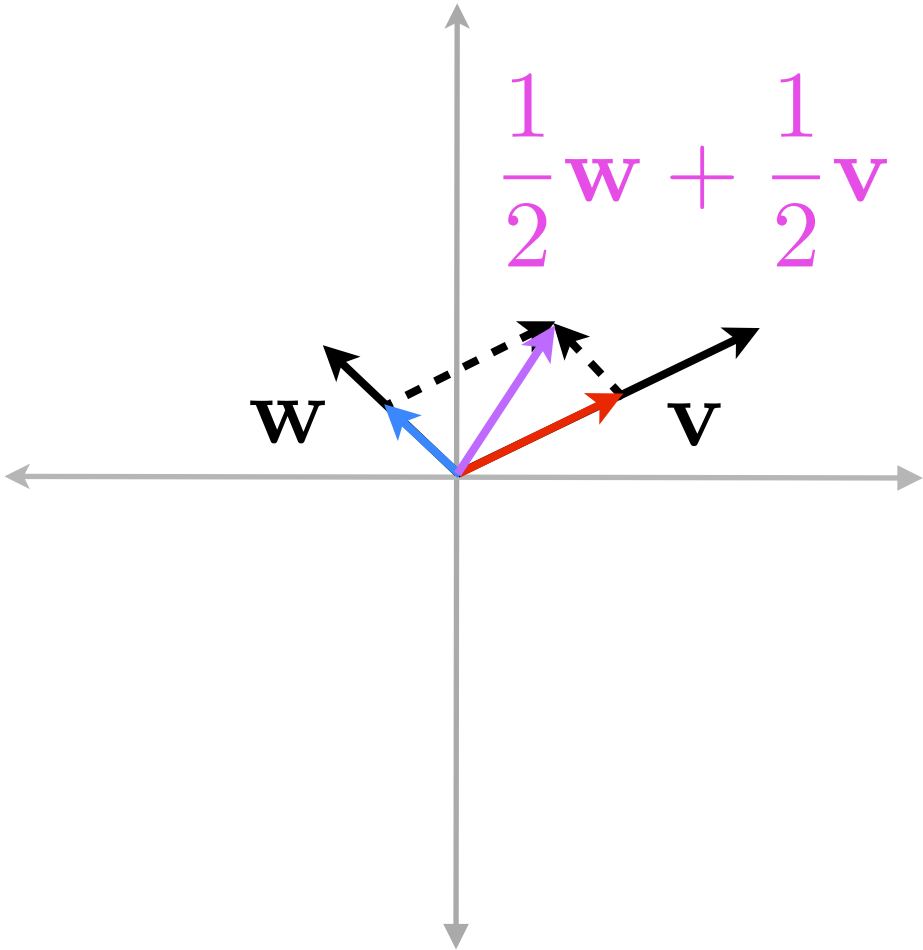
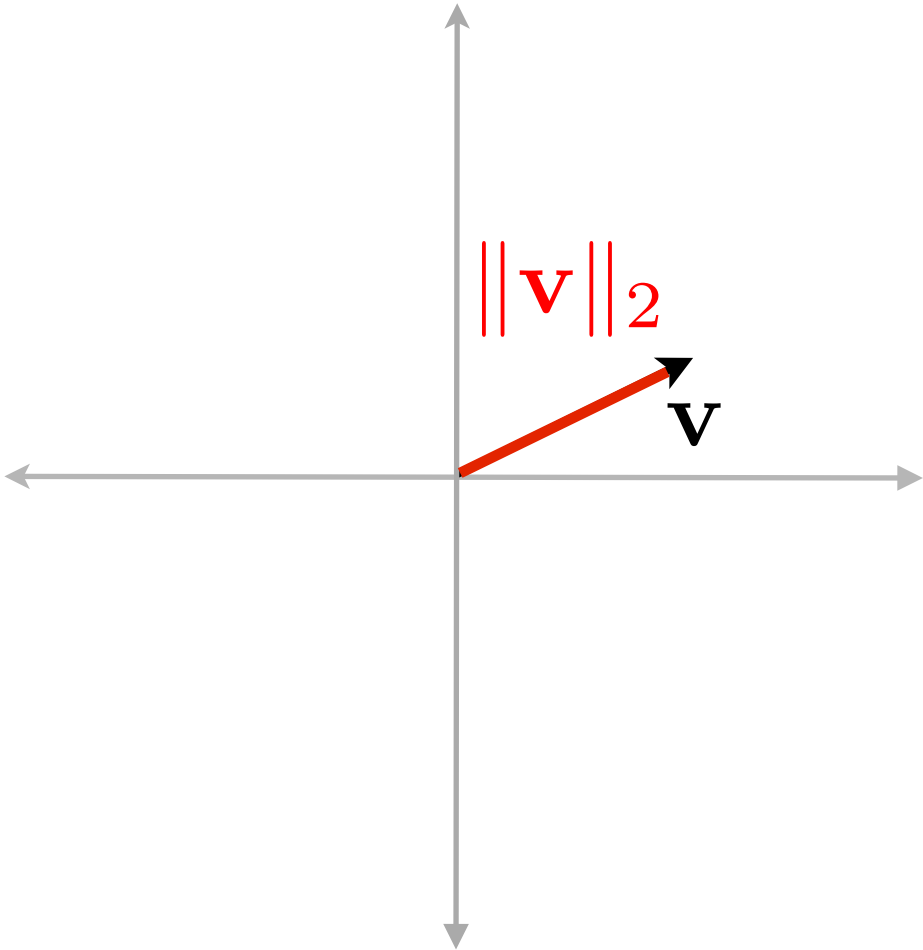
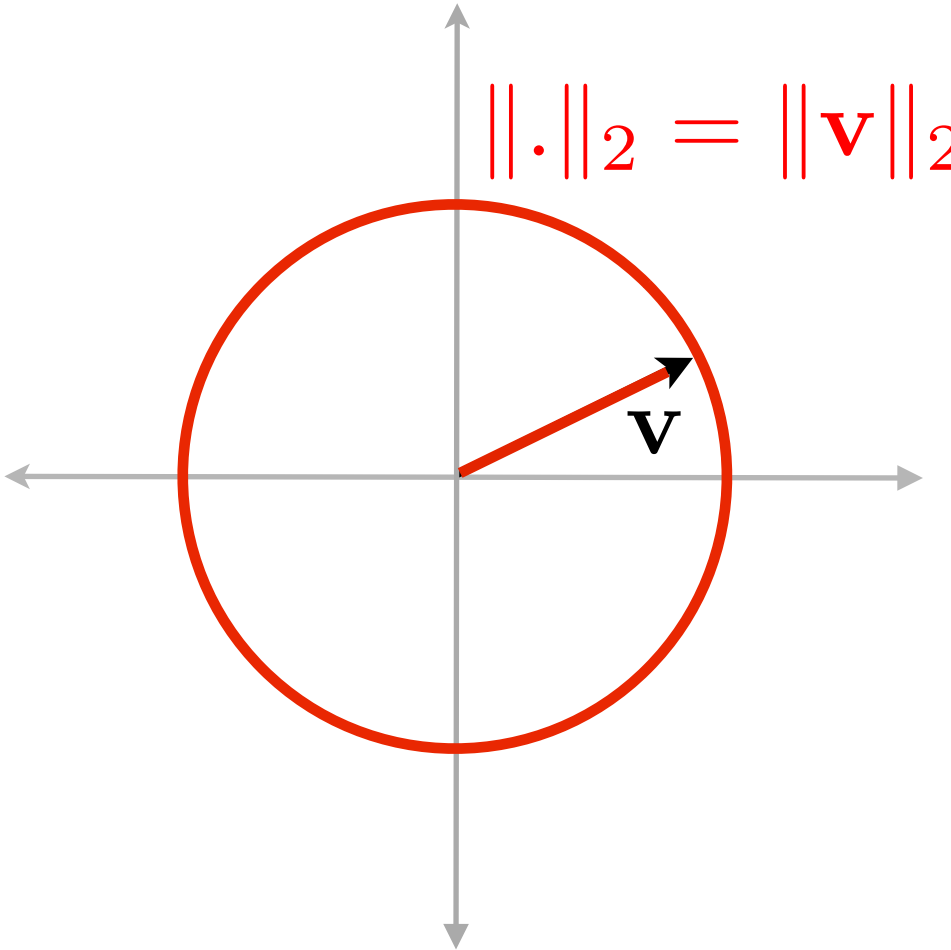

v

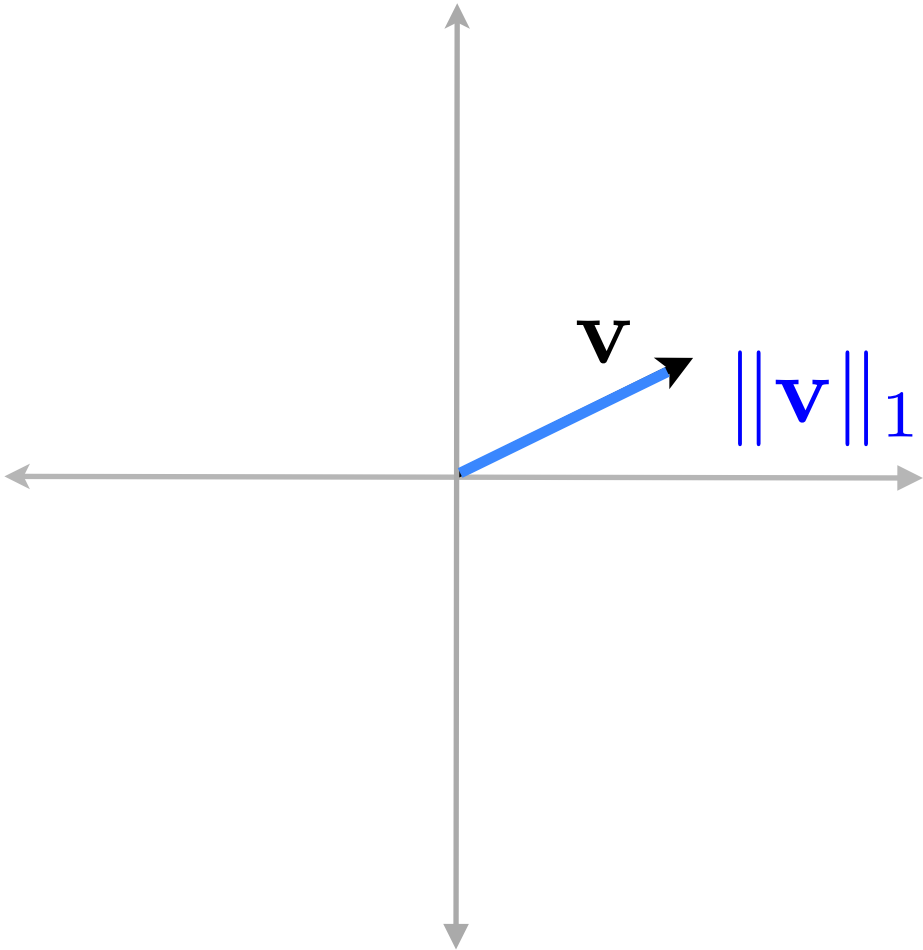

$a\mathbf{v}$
$\mathbf{v}$

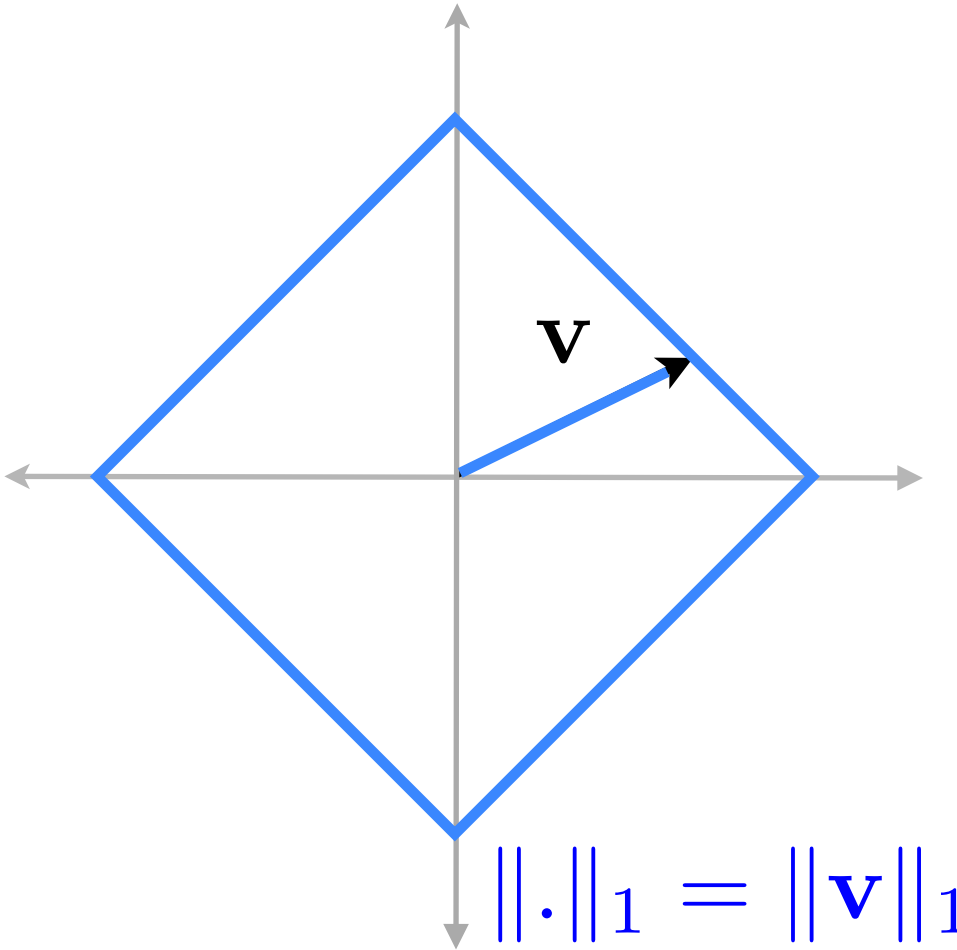

$\mathbf{v} + \mathbf{w}$
$\mathbf{w}$
$\mathbf{v}$

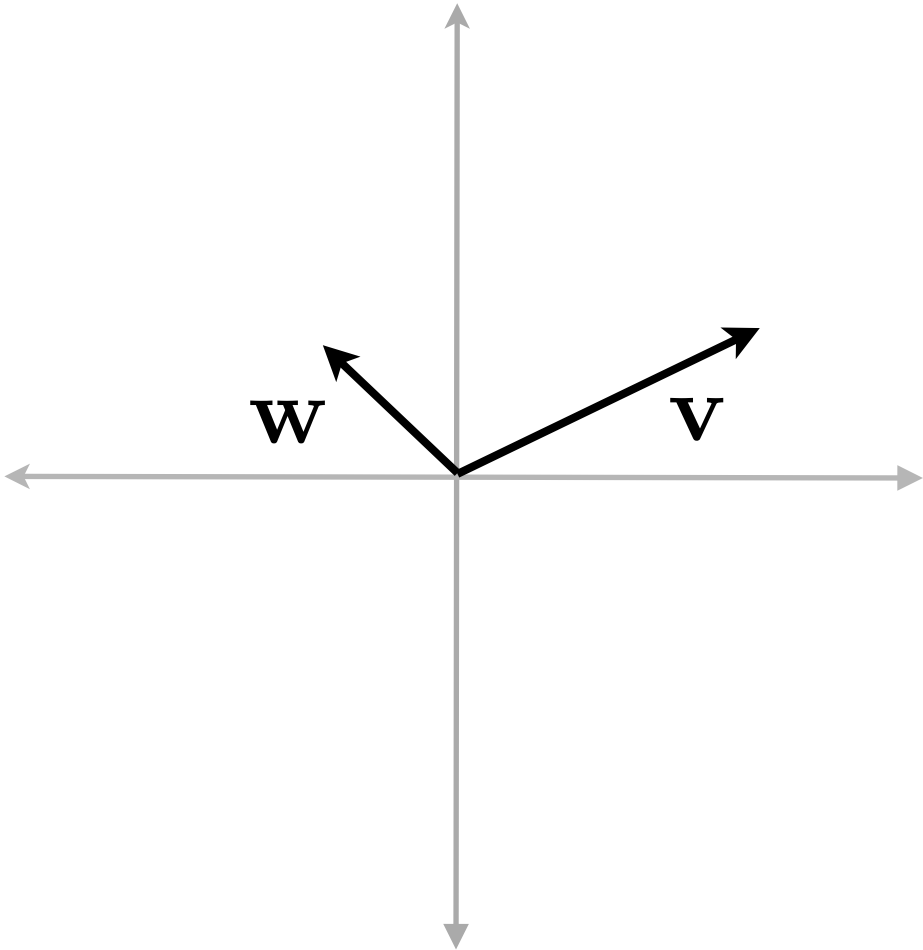

$\frac{1}{2}\mathbf{w} + \frac{1}{2}\mathbf{v}$
$\mathbf{w}$
$\mathbf{v}$

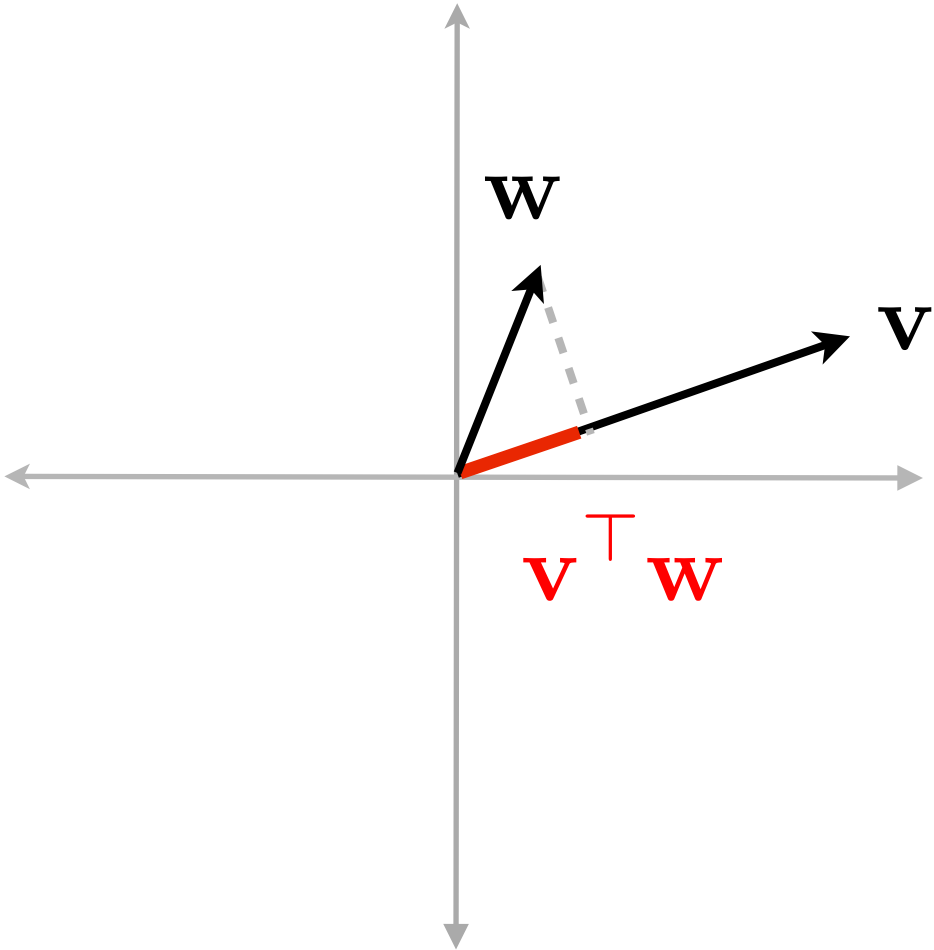

$\|\mathbf{v}\|_2$
$\mathbf{v}$

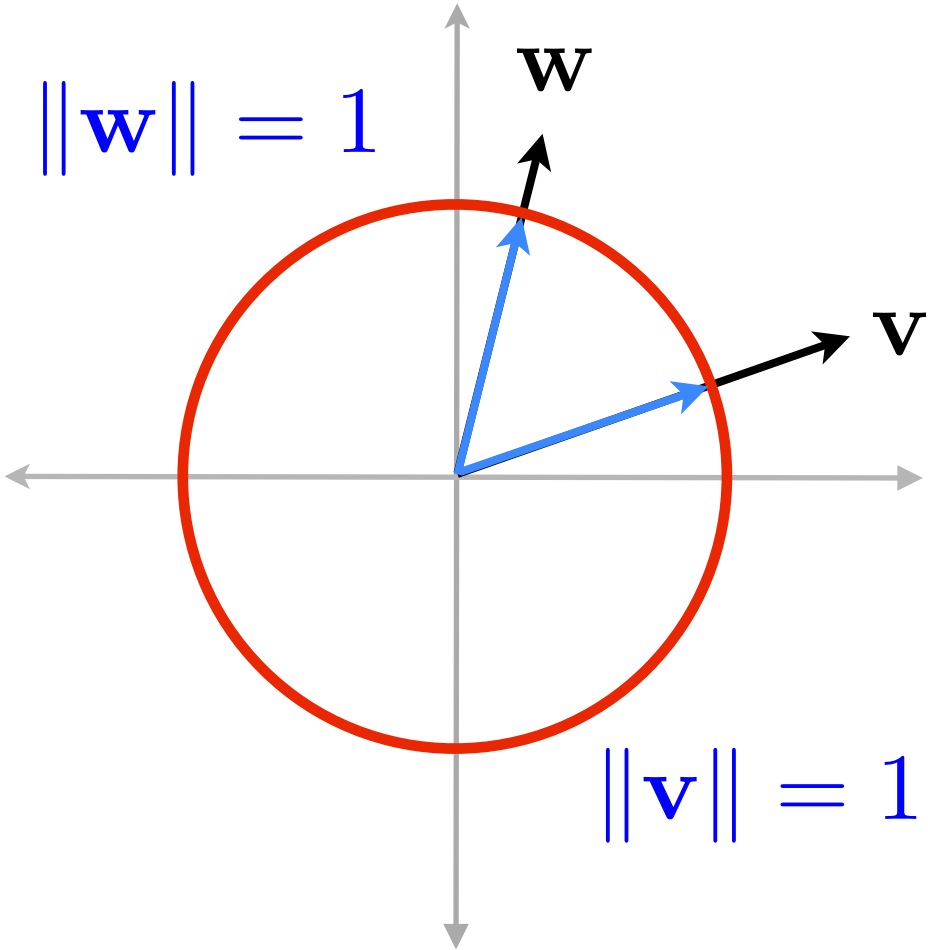

$\|\cdot\|_2 = \|\mathbf{v}\|_2$
$\mathbf{v}$

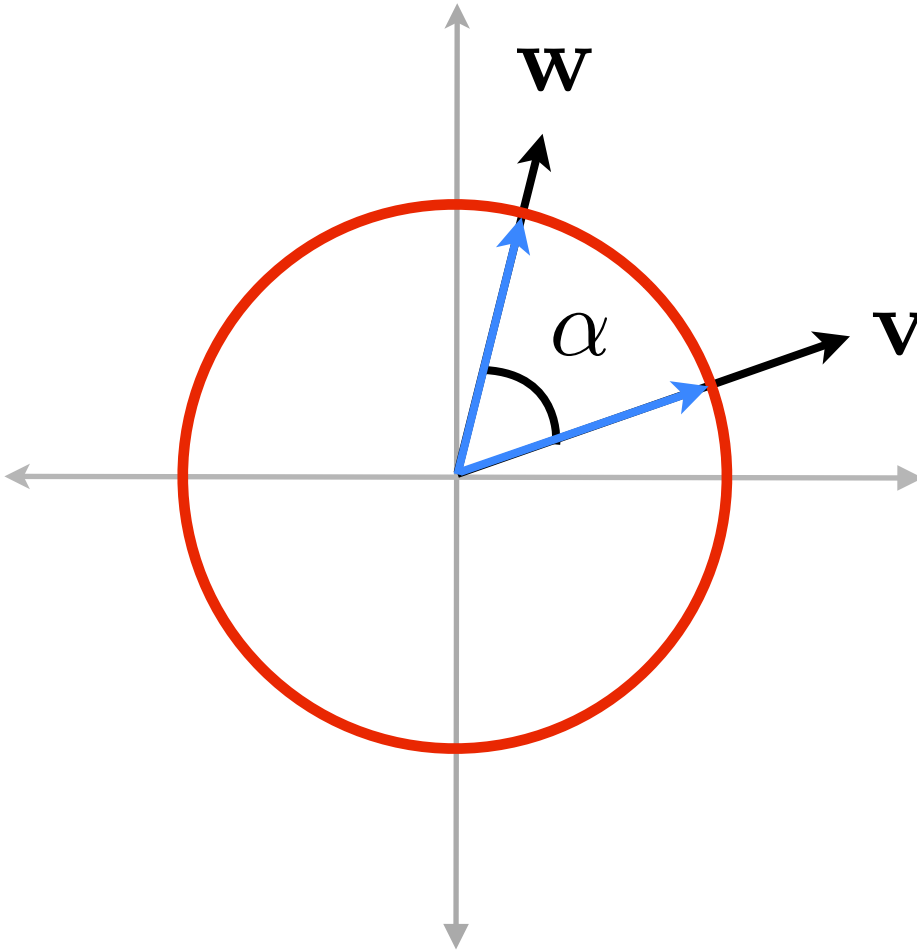

$\mathbf{v}$
$\|\mathbf{v}\|_1$

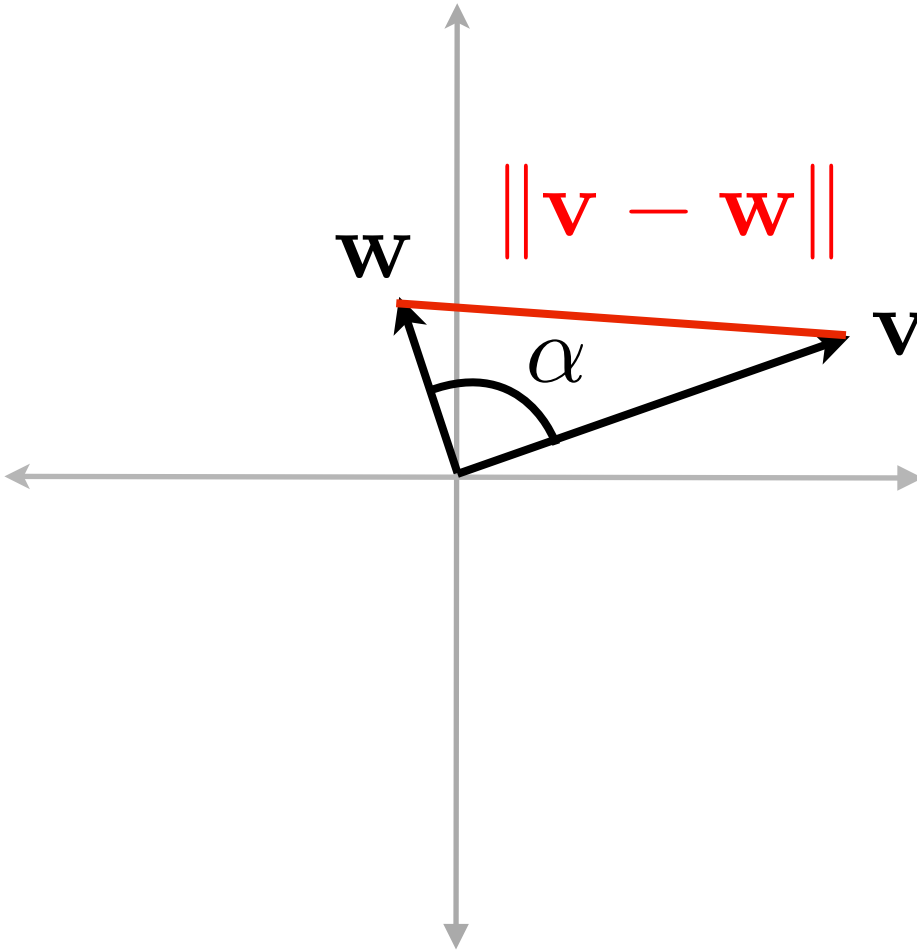

$\mathbf{v}$
$\|\cdot\|_1 = \|\mathbf{v}\|_1$

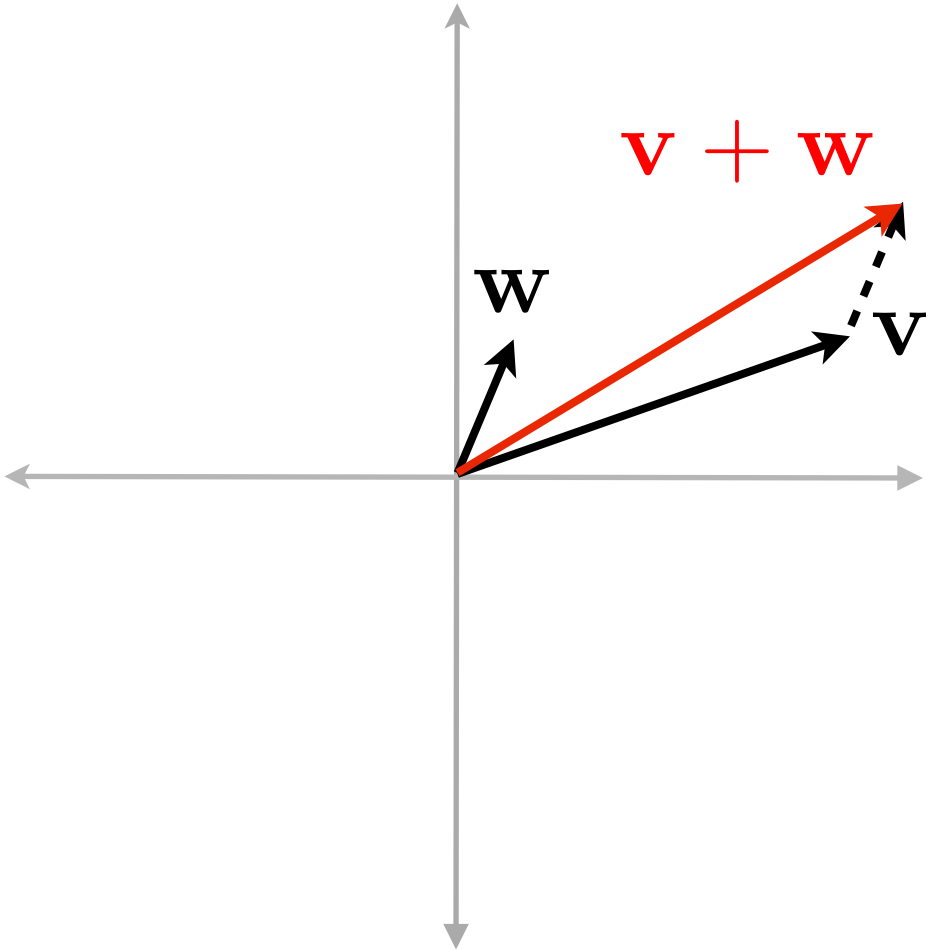

w
v

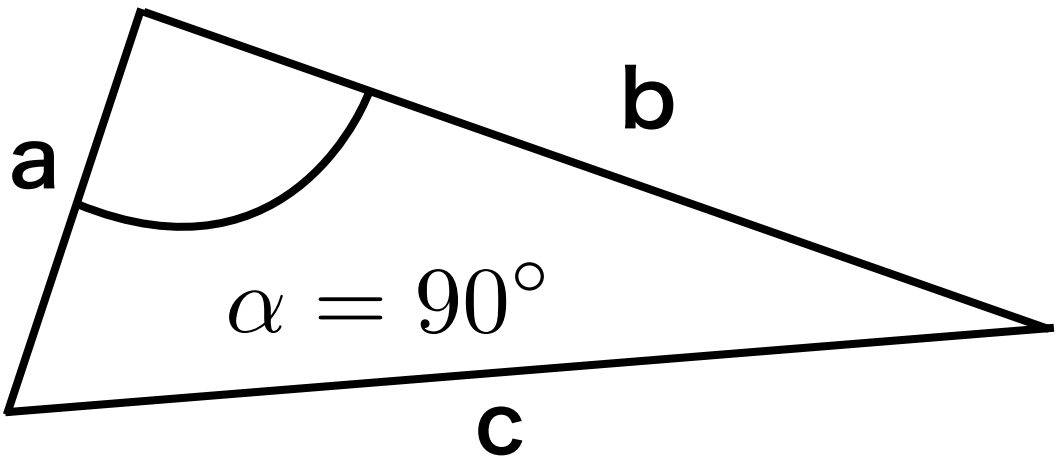

$\mathbf{w}$
$\mathbf{v}$
$\mathbf{v}^\top \mathbf{w}$


$\|\mathbf{w}\| = 1$
$\mathbf{w}$
$\mathbf{v}$
$\|\mathbf{v}\| = 1$


w
v
α


$\|\mathbf{v} - \mathbf{w}\|$
$\mathbf{w}$
$\mathbf{v}$
α


$\mathbf{v} + \mathbf{w}$
$\mathbf{w}$
$\mathbf{v}$


a
b
c
$\alpha = 90°$